

CLINICAL STUDY PROTOCOL

Protocol Number:

HTN001

Study Title:

A Randomized, Double-Blind, Placebo-Controlled, Parallel-Group, Phase II Study to Evaluate the Efficacy and Safety of Drug A in Adult Patients with Essential Hypertension

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Clinical SAS Portfolio Project

1. Introduction

1.1 Background

Hypertension is a chronic medical condition characterized by persistently elevated arterial blood pressure and is one of the leading modifiable risk factors for cardiovascular morbidity and mortality worldwide. Essential (primary) hypertension accounts for approximately 90–95% of hypertension cases and develops without an identifiable secondary cause. The remaining cases are classified as secondary hypertension and result from underlying medical conditions such as renal disease, endocrine disorders, or the use of certain medications.

The prevalence of hypertension continues to increase due to population aging, urbanization, sedentary lifestyles, obesity, unhealthy dietary habits, and excessive sodium intake. Although effective treatment options are available, a substantial proportion of patients remain undiagnosed, untreated, or inadequately controlled, placing them at increased risk of long-term complications.

Uncontrolled hypertension is associated with serious cardiovascular and renal outcomes, including stroke, myocardial infarction, heart failure, chronic kidney disease, peripheral arterial disease, and premature death. Early diagnosis and effective blood pressure management are therefore essential to reduce disease burden and improve long-term clinical outcomes.

Given its high prevalence and significant impact on global public health, hypertension remains an important therapeutic area requiring the development of safe and effective treatment options that provide sustained blood pressure control and improve patient adherence.

1.2 Current Treatment

Current management of essential hypertension includes both lifestyle modification and pharmacological therapy. Lifestyle interventions such as dietary sodium restriction, regular physical activity, weight reduction, smoking cessation, moderation of alcohol intake, and consumption of a balanced diet are recommended for all patients.

Pharmacological treatment commonly includes angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor blockers (ARBs), calcium channel blockers, thiazide or thiazide-like diuretics, and beta-blockers when clinically appropriate. Treatment selection is based on patient characteristics, comorbidities, and current clinical practice guidelines.

Despite the availability of multiple antihypertensive medications, many patients fail to achieve recommended blood pressure targets because of inadequate response, adverse events, poor treatment adherence, or the need for combination therapy. These limitations

highlight the need for additional therapeutic options with improved efficacy, safety, and patient convenience.

1.3 Investigational Drug

Drug A is a novel, orally administered investigational antihypertensive agent being developed for the treatment of adult patients with essential hypertension. Drug A is designed to provide sustained blood pressure reduction through a selective mechanism of action that targets pathways involved in vascular regulation while minimizing undesirable systemic effects.

Preclinical studies and early clinical investigations have demonstrated promising antihypertensive activity and an acceptable safety profile. The once-daily dosing regimen of Drug A may improve treatment adherence while maintaining consistent blood pressure control throughout the dosing interval.

As Drug A remains investigational, its efficacy and safety have not yet been established for routine clinical use and require confirmation in adequately designed clinical studies.

1.4 Study Rationale

Although several antihypertensive therapies are currently available, many patients continue to experience inadequate blood pressure control or treatment-limiting adverse events. There remains a clinical need for new therapies capable of providing effective, sustained blood pressure reduction with a favorable safety and tolerability profile.

Drug A has demonstrated encouraging results in nonclinical and early-phase clinical studies, supporting further evaluation in patients with essential hypertension. This Phase II clinical study has been designed to assess the efficacy and safety of Drug A compared with placebo in adult patients with essential hypertension.

The results of this study will contribute to the assessment of the therapeutic potential of Drug A and provide evidence to support future clinical development.

2. Study Objectives

2.1 Primary Objective

To evaluate the efficacy of Drug A compared with placebo in reducing systolic blood pressure in adult patients with essential hypertension after 12 weeks of treatment.

2.2 Secondary Objectives

- To evaluate the effect of Drug A on diastolic blood pressure.
- To assess the proportion of patients achieving target blood pressure at Week 12.
- To evaluate the safety and tolerability of Drug A throughout the study.
- To assess patient compliance with the study treatment.

2.3 Exploratory Objectives

- To explore changes in selected cardiovascular biomarkers following treatment with Drug A.
- To evaluate changes in patient-reported quality of life during the study.
- To explore the relationship between baseline patient characteristics and treatment response.

3.Study Design

3.1 Study Overview

This is a Phase II, multicenter, randomized, double-blind, placebo-controlled, parallel-group clinical study designed to evaluate the efficacy and safety of Drug A in adult patients with essential hypertension. Approximately 120 eligible subjects will be enrolled and randomly assigned in a 1:1 ratio to receive either Drug A 50 mg or matching placebo once daily for a treatment period of 12 weeks.

The study consists of a screening period, a baseline visit, scheduled treatment visits, and an end-of-study visit. Clinical assessments, including vital signs, laboratory evaluations, adverse event monitoring, concomitant medication review, and treatment compliance assessments, will be performed according to the study schedule.

The primary objective of the study is to evaluate the reduction in systolic blood pressure from baseline after 12 weeks of treatment. Secondary objectives include evaluation of diastolic blood pressure, safety profile, laboratory parameters, and treatment tolerability.

3.2 Study Design Summary

Study Title	A Phase II Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Safety and Efficacy of Drug A in Adult Patients with Essential Hypertension
Protocol Number	HTN001
Phase	II
Study Design	Randomized, Double-Blind, Placebo-Controlled, Parallel-Group
Therapeutic Area	Hypertension
Study Population	Adults with Essential Hypertension
Number of Subjects	Approximately 120
Treatment Arms	Drug A 50 mg, Placebo
Randomization Ratio	1:1
Treatment Duration	12 weeks
Screening Period	Up to 14 days
Primary Endpoint	Change from Baseline in Systolic Blood Pressure
Secondary Endpoint	Change in Diastolic Blood Pressure, Safety assessments, Laboratory Parameters, Adverse Events
Statistical Analysis Population	Safety Population, Intent to treat Population

3.4 Randomization

Eligible subjects will be randomly assigned in a 1:1 ratio to receive either Drug A 50 mg or placebo. Randomization will be performed using a computer-generated randomization

schedule prepared prior to study initiation. Each enrolled subject will receive a unique subject identification number and a corresponding treatment assignment according to the randomization schedule.

3.5 Blinding

This study will be conducted using a double-blind design. Subjects, investigators, study personnel, and the sponsor's study team will remain blinded to treatment assignment throughout the study. Drug A and placebo will be identical in appearance, packaging, and labelling to maintain the integrity of the blind. Emergency unblinding will be permitted only when knowledge of the treatment assignment is essential for safety.

3.6 Treatment Groups

Treatment Group	Treatment	Dose	Frequency
Group A	Drug A	50 mg	Once Daily
Group B	Placebo	Matching Tablet	Once Daily

3.7 Study Duration

Study Period	Duration
Screening	Up to 14 Days
Treatment	12 Weeks
Follow-up	7 Days
Total Participation	Approximately 15 Weeks

3.8 Study Visits

Visits	Weeks	Description
Screening	-2	Eligibility Assessment
Baseline	0	Randomization and First Dose
Visit 2	4	Efficacy and Safety Assessment
Visit 3	8	Efficacy and Safety Assessment
End of Treatment	12	Final Assessment
Follow-up	13	Safety Follow-up

3.9 Schedule of Assessments

4. Study Population

This study will enroll adult male and female subjects with essential hypertension who meet all protocol-defined eligibility criteria. Subject selection is intended to ensure that the study population is appropriate for evaluating the efficacy and safety of Drug A while minimizing potential risks to participants. Eligibility will be determined during the screening period through medical history, physical examination, vital signs, laboratory assessments, and other protocol-required evaluations.

4.1 Inclusion Criteria

Subjects must meet all of the following criteria to be eligible for enrollment in this study:

1. Male or female subjects aged 18 to 65 years, inclusive, at the time of signing the informed consent form.
2. A confirmed diagnosis of essential (primary) hypertension according to the investigator's clinical judgment.
3. Mean seated systolic blood pressure (SBP) between 140 and 179 mmHg and/or diastolic blood pressure (DBP) between 90 and 109 mmHg at the Screening Visit and confirmed at the Baseline Visit.
4. Body Mass Index (BMI) between 18.5 and 40.0 kg/m², inclusive.
5. Able and willing to provide written informed consent before the performance of any study-specific procedures.
6. Willing and able to comply with all study visits, study procedures, and protocol requirements throughout the study duration.
7. Clinical laboratory test results, medical history, physical examination, and 12-lead ECG considered acceptable by the investigator for study participation.
8. Women of childbearing potential must have a negative pregnancy test at Screening and Baseline and must agree to use a highly effective method of contraception throughout the study and for the protocol-defined period after the last dose of study treatment.
9. Male subjects with partners of childbearing potential must agree to use appropriate contraception during the study and for the protocol-defined period after the last dose of study treatment, if required by local regulations.
10. In the opinion of the investigator, the subject is medically suitable for participation and is expected to complete all required study procedures.

4.2 Exclusion Criteria

Subjects meeting any of the following criteria will be excluded from participation in this study:

1. Secondary hypertension resulting from an identifiable underlying medical condition.
2. Severe or uncontrolled hypertension, defined as a mean seated systolic blood pressure ≥ 180 mmHg or diastolic blood pressure ≥ 110 mmHg at Screening or Baseline.
3. History of hypertensive emergency or hypertensive crisis within 6 months before Screening.
4. History of clinically significant cardiovascular disease, including myocardial infarction, stroke, unstable angina, or hospitalization for heart failure within 6 months prior to Screening.
5. Clinically significant renal impairment, hepatic disease, or other uncontrolled systemic illness that, in the opinion of the investigator, may interfere with study participation or interpretation of study results.
6. Clinically significant abnormal laboratory values or electrocardiogram (ECG) findings that may place the subject at increased risk or affect study outcomes.
7. Known hypersensitivity or allergy to Drug A, its excipients, or related compounds.
8. Pregnant or breastfeeding women, or women planning to become pregnant during the study.
9. Use of any investigational drug or participation in another clinical trial within 30 days or five half-lives of the investigational product (whichever is longer) before Screening.
10. History of alcohol or substance abuse within the previous 12 months that may interfere with study participation or compliance.
11. Presence of any medical, psychiatric, or social condition that, in the opinion of the investigator, may compromise subject safety, protocol compliance, or interpretation of study results.
12. Any condition that, in the judgment of the investigator or sponsor, makes the subject unsuitable for participation in this clinical study.

4.3 Withdrawal Criteria

Subjects may voluntarily withdraw from the study at any time without providing a reason and without prejudice to their future medical care. The investigator may also withdraw a

subject from the study if continued participation is considered not to be in the subject's best interest.

A subject may be withdrawn from the study for any of the following reasons:

1. Withdrawal of informed consent.
2. Occurrence of a serious adverse event (SAE) or any adverse event that, in the opinion of the investigator, requires discontinuation of the study treatment.
3. Development of a medical condition that makes continued participation unsafe.
4. Pregnancy during the study.
5. Major protocol deviation that may affect subject safety or compromise the integrity of the study.
6. Poor compliance with study medication or study procedures.
7. Loss to follow-up despite reasonable efforts to contact the subject.
8. Use of prohibited concomitant medication that may interfere with the evaluation of the investigational product.
9. Decision by the investigator or sponsor to discontinue the subject for safety, administrative, or other justified reasons.

Whenever possible, subjects who discontinue study treatment prematurely should complete an Early Termination Visit, including all protocol-specified safety and efficacy assessments. The reason for study discontinuation will be documented in the study records and electronic Case Report Form (eCRF).

4.4 Replacement of Subjects

Subjects who discontinue the study before receiving the first dose of study treatment may be replaced at the discretion of the sponsor to ensure that the planned sample size is achieved.

Subjects who discontinue the study after receiving at least one dose of the investigational product will generally not be replaced. These subjects will be included in the appropriate analysis populations according to the Statistical Analysis Plan (SAP).

All subject discontinuations and replacements, if applicable, will be documented in the study records and the electronic Case Report Form (eCRF). The reason for discontinuation will be recorded and summarized in the final clinical study report.

5. Study Treatment

5.1 Investigational Product

The investigational product, Drug A, is an orally administered antihypertensive agent being evaluated for the treatment of adult subjects with essential hypertension. Drug A will be supplied as 50 mg film-coated tablets for oral administration.

Drug A is intended for investigational use only and will be administered in accordance with the study protocol. The investigational product will be packaged, labeled, and stored according to applicable regulatory requirements and Good Clinical Practice (GCP).

Matching placebo tablets identical in appearance, packaging, labeling, and administration schedule will be used to maintain the double-blind design of the study.

5.2 Dosage and Administration

Eligible subjects randomized to the active treatment group will receive Drug A 50 mg orally once daily for 12 weeks. Subjects randomized to the control group will receive matching placebo orally once daily for the same treatment duration.

Study medication should be taken at approximately the same time each day with water and may be administered with or without food, unless otherwise instructed by the investigator.

Missed doses should not be replaced. Subjects should take the next scheduled dose at the usual dosing time. Double dosing to compensate for a missed dose is not permitted.

Dose modification is not permitted during the study unless required for subject safety or permanent discontinuation of study treatment.

5.3 Treatment Compliance

Treatment compliance will be assessed throughout the study to ensure that subjects receive the investigational product as specified in the protocol.

Subjects will be instructed to take the assigned study medication once daily for the duration of the treatment period. At each scheduled study visit, treatment compliance will be evaluated by review of subject dosing records and by counting returned study medication.

The investigator or designated study personnel will counsel subjects with poor compliance and document any significant non-compliance in the study records and the electronic Case Report Form (eCRF).

Subjects who demonstrate persistent non-compliance may be withdrawn from the study if continued participation is considered likely to compromise subject safety or the integrity of the study.

5.4 Concomitant Medications

All concomitant medications taken by the subject from the time of signing the informed consent form until completion of the study will be recorded in the electronic Case Report Form (eCRF).

The medication name, indication, dose, route of administration, frequency, and start and stop dates will be documented. Concomitant medications considered necessary for the subject's medical care may be continued during the study unless prohibited by the protocol.

5.5 Prohibited Medications

The use of medications that may interfere with the evaluation of the efficacy or safety of Drug A should be avoided during the study.

Prohibited medications include, but are not limited to:

- Other investigational medicinal products.
- Non-protocol antihypertensive medications unless required for emergency management.
- Medications known to significantly affect blood pressure unless approved by the sponsor.
- Any medication specifically prohibited by the Investigator's Brochure or study protocol.

If a prohibited medication is required for the medical management of a subject, the investigator will determine whether continuation in the study is appropriate and document the decision accordingly.

5.6 Study Drug Storage and Accountability

The investigational product and matching placebo will be stored under the conditions specified on the product label and in accordance with applicable regulatory requirements and institutional procedures.

Access to study medication will be restricted to authorized study personnel. Drug accountability records will be maintained throughout the study to document receipt, dispensing, return, and final disposition of all study medication.

At each study visit, the quantity of study medication dispensed and returned will be recorded. Any discrepancies identified during drug accountability reconciliation will be investigated and documented.

Unused or partially used study medication will be returned or destroyed according to sponsor instructions and applicable regulatory requirements.

6. Study Procedures

6.1 Screening Procedures

The Screening Visit will be performed within 14 days prior to randomization to determine subject eligibility. Written informed consent must be obtained before the initiation of any study-specific procedures.

The following assessments will be performed during the Screening Visit, as applicable:

- Obtain written informed consent.
- Record demographic information and relevant medical history.
- Review prior and concomitant medications.
- Perform a complete physical examination.
- Measure height, weight, and body mass index (BMI).
- Measure seated systolic and diastolic blood pressure.
- Measure pulse rate and body temperature.
- Perform a 12-lead electrocardiogram (ECG).
- Collect blood and urine samples for clinical laboratory evaluations.
- Assess eligibility according to the protocol-defined inclusion and exclusion criteria.
- Assign a screening identification number

6.2 Baseline (Day 1)

Subjects who satisfy all eligibility criteria will return for the Baseline Visit (Day 1).

The following procedures will be performed:

- Confirm subject eligibility.
- Confirm inclusion and exclusion criteria.
- Perform randomization.
- Assign the randomized treatment.
- Dispense study medication.
- Record baseline vital signs.
- Record baseline laboratory results, if required.
- Review concomitant medications.
- Assess adverse events since Screening.
- Provide study medication administration instructions.

- Record treatment compliance instructions.

6.7 Schedule of Assessments

The Schedule of Assessments summarizes the procedures and evaluations to be performed at each study visit. Unless otherwise specified, all assessments should be completed according to the visit schedule defined in this protocol. Additional assessments may be performed at the discretion of the investigator if clinically indicated.

Assessment	Screening (-14 to -1)	Baseline (Day 1)	Week 4	Week 8	Week 12 (End of Treatment)	Follow-up (7 Days)
Informed Consent						
Inclusion/Exclusion Criteria						
Demographics						
Medical History						
Physical Examination						
Height						
Weight						
Blood Pressure						
Pulse Rate						
Body Temperature						
12-Lead ECG						
Clinical Laboratory Tests						
Pregnancy Test*						
Randomization						
Study Drug Dispensing						
Study Drug Accountability						
Treatment Compliance						
Concomitant Medications						
Adverse Event Assessment						
End of Study Procedures						
Follow-up Safety Assessment						